

Low Rank Parareal: a new combination of Parareal with dynamical low rank approximation

Benjamin Carrel

Joint work with M.J.Gander and B.Vandereycken



**UNIVERSITÉ
DE GENÈVE**

FACULTÉ DES SCIENCES

Section de Mathématiques

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Schedule

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- 3 Low rank Parareal: a practical formulation
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Introduction: setup

Consider an initial value problem

$$\begin{aligned}\dot{X}(t) &= F(X(t)), \quad t \in [0, T] \\ X(0) &= X_0,\end{aligned}$$

where $X(t) \in \mathbb{R}^{m \times n}$ for every time t . This is called a **matrix** differential equation. Suppose the solution admits good low-rank approximation,

$$X \approx Y \in \mathcal{M}_r = \{A \in \mathbb{R}^{m \times n} | \text{rank}(A) = r\}.$$

Question: can we solve efficiently this problem in parallel in time?



Introduction: Parareal

Definition (Parareal algorithm ([Lions et al., 2001]))

Iterate over n and k ,

$$\text{(Initial value)} \quad U_0^k = X_0, \quad k = 0, \dots, N$$

$$\text{(Initial iteration)} \quad U_{n+1}^0 = \mathcal{G}^h(U_n^0), \quad n = 0, \dots, N$$

$$\text{(Iteration)} \quad U_{n+1}^{k+1} = \mathcal{F}^h(U_n^k) + \mathcal{G}^h(U_n^{k+1}) - \mathcal{G}^h(U_n^k), \quad k, n = 0, \dots, N$$

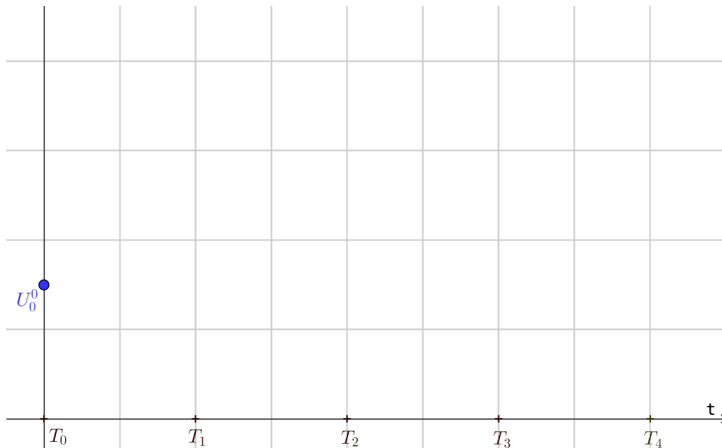
where $\mathcal{F}$ represents the fine solver and $\mathcal{G}$ the coarse solver.

- Easy to implement.
- Good on parabolic problems.
- Finite step convergence.
- Bad on hyperbolic problems.



Introduction: Parareal

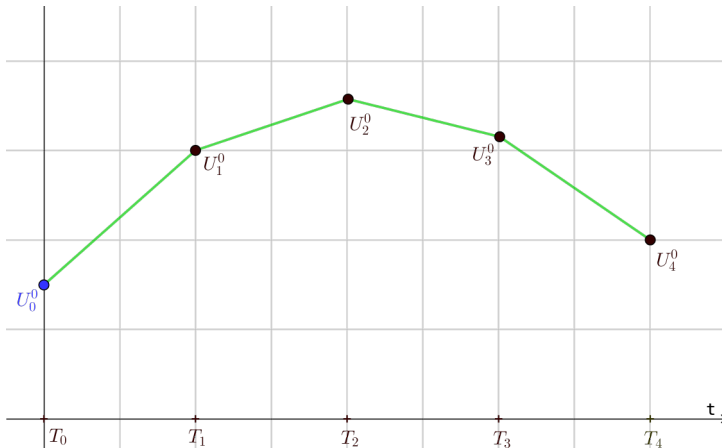
Time discretization: $[0, T] = [T_0, T_1] \cup [T_1, T_2] \cup \dots \cup [T_{N-1}, T_N]$.





Introduction: Parareal

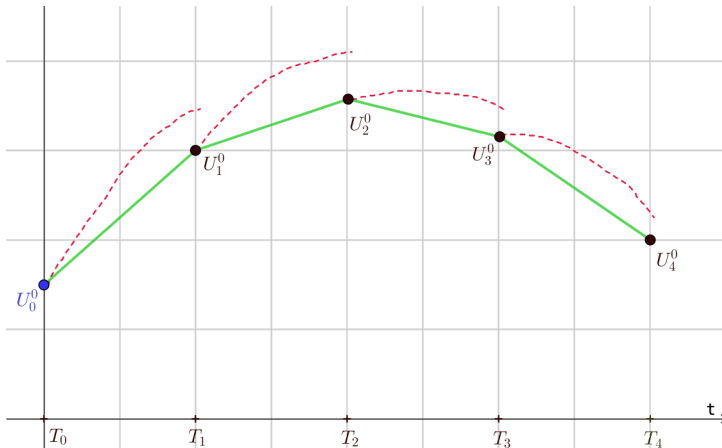
Initial iteration: apply the coarse solver iteratively.





Introduction: Parareal

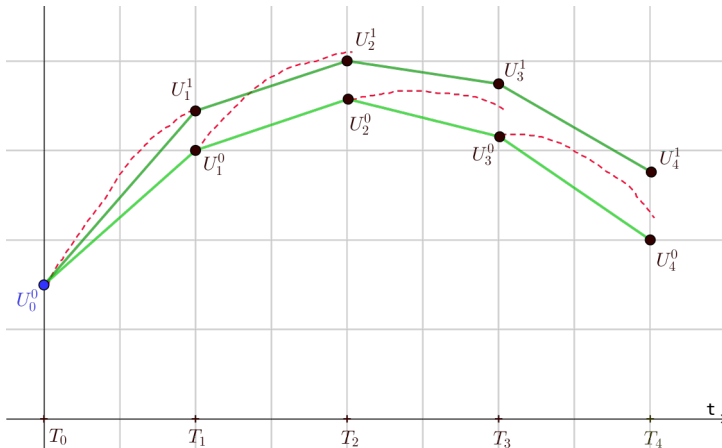
Compute in parallel: $\mathcal{F}^h(U_n^0)$.





Introduction: Parareal

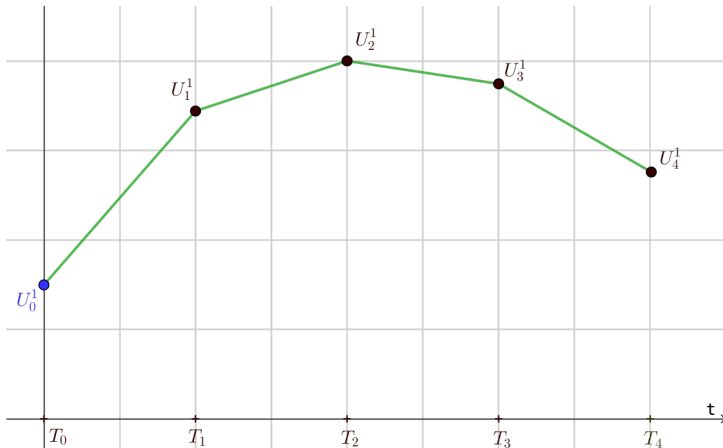
Propagate sequentially using the coarse solver.





Introduction: Parareal

Repeat the process...





Introduction: DLRA

GOAL: for every t , find $Y(t) \in \mathcal{M}_r$ such that $\|X(t) - Y(t)\| \stackrel{!}{=} \min$.

Original problem

$$\begin{aligned}\dot{X}(t) &= F(X(t)), \quad t \in [0, T] \\ X(0) &= X_0.\end{aligned}$$

DLRA ([Koch and Lubich, 2007])

$$\begin{aligned}\dot{Y}(t) &= \mathcal{P}_{Y(t)} [F(Y(t))], \quad t \in [0, T] \\ Y(0) &= Y_0 \in \mathcal{M}_r.\end{aligned}$$

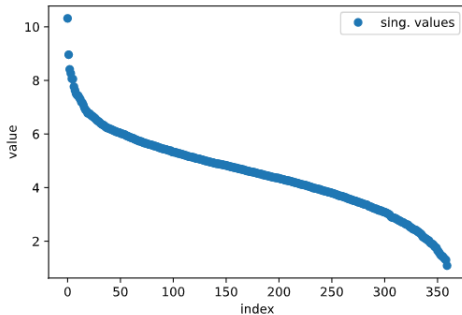
Motivations

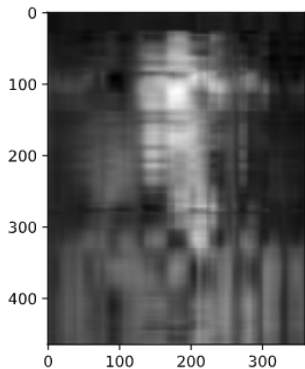
- $\text{rank}(Y(t)) = r \ll m, n \rightarrow$ efficient storage and less computations.
- (Under conditions) DLRA is accurate: $Y(t) \approx X(t)$. [Koch and Lubich, 2007]
[Kieri et al.,]
- Strong mathematical construction: Smooth manifolds.



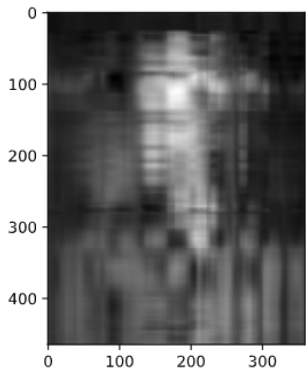
Introduction: DLRA

Let's do a low-rank approximation of a famous Swiss mathematician.





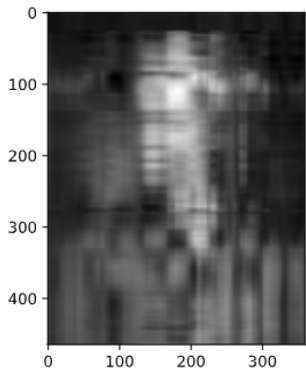
- (a) Rank 5 approximation.
Relative error: 23.5%
Compression : $72\times$ smaller.



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(b) Rank 15 approximation.
Relative error: 14.5%
Compression : $24\times$ smaller.



(c) Rank 50 approximation.
Relative error: 7.8%
Compression : $7.2\times$ smaller.



Idealized method: definition

Original problem

$$\begin{aligned}\dot{X}(t) &= F(X(t)), \quad t \in [0, T] \\ X(0) &= X_0.\end{aligned}$$

DLRA ([Koch and Lubich, 2007])

$$\begin{aligned}\dot{Y}(t) &= \mathcal{P}_{Y(t)} [F(Y(t))], \quad t \in [0, T] \\ Y(0) &= Y_0 \in \mathcal{M}_r.\end{aligned}$$

Definition (Low rank Parareal, idealized formulation [C., Gander, Vandereycken])

For two ranks q, r such that $q < r$, we iterate

$$\text{(Initial value)} \quad U_0^k = \mathcal{T}_q(X_0)$$

$$\text{(Initial iteration)} \quad U_{n+1}^0 = \phi^h \circ \mathcal{T}_q(U_n^0)$$

$$\text{(Iteration)} \quad U_{n+1}^{k+1} = \phi^h \circ \mathcal{T}_r(U_n^k) + \phi^h \circ \mathcal{T}_q(U_n^{k+1}) - \phi^h \circ \mathcal{T}_q(U_n^k)$$

where ϕ^h is the flow of the original problem, and $\mathcal{T}_q$ is the truncation to rank q .

GOAL: Understand the role of the two ranks q and r .



Idealized method: analysis

Consider an affine problem

$$F(X(t)) = \mathcal{A}X(t) + \mathcal{B}.$$

Let's study the iteration on the error,

$$\begin{aligned} E_{n+1}^{k+1} &= X(t_{n+1}) - U_{n+1}^{k+1} \\ &= X(t_{n+1}) - \phi^h(U_n^k) - \phi^h \circ \mathcal{T}_q(U_n^{k+1}) + \phi^h \circ \mathcal{T}_q(U_n^k) \\ &= \phi^h(X(t_n)) - \phi^h \circ \mathcal{T}_q(X(t_n)) - \phi^h(U_n^k) + \phi^h \circ \mathcal{T}_q(U_n^k) \\ &\quad + \phi^h \circ \mathcal{T}_q(X(t_n)) - \phi^h \circ \mathcal{T}_q(U_n^{k+1}) \\ &= \phi^h \circ \mathcal{T}_q^\perp(X(t_n)) - \phi^h \circ \mathcal{T}_q^\perp(U_n^k) + \phi^h \circ \mathcal{T}_q(X(t_n)) - \phi^h \circ \mathcal{T}_q(U_n^{k+1}) \end{aligned}$$

$$\implies \|E_{n+1}^{k+1}\| \leq \alpha \|E_n^k\| + \beta \|E_n^{k+1}\|,$$

where α and β are the Lipschitz constants of $\phi^h \circ \mathcal{T}_q^\perp$ and $\phi^h \circ \mathcal{T}_q$.



Idealized method: analysis

$$\left\| E_{n+1}^{k+1} \right\| \leq \alpha \left\| E_n^k \right\| + \beta \left\| E_n^{k+1} \right\|$$

Let's estimate α , the Lipschitz constant of $\phi^h \circ \mathcal{T}_q^\perp$.

$$\alpha = \left\| D \left(\phi^h \circ \mathcal{T}_q^\perp \right) \right\|_{op} \leq \left\| D\phi^h \right\|_{op} \left\| D\mathcal{T}_q^\perp \right\|_{op}.$$

First,

$$\phi^h(X) = e^{\mathcal{A}h}X + \int_0^t e^{\mathcal{A}s}\mathcal{B}ds \implies \left\| D\phi^h \right\|_{op} \leq e^{\lambda h},$$

and by [Breiding and Vannieuwenhoven, 2021], $\left\| D\mathcal{T}_q^\perp \right\|_{op} \leq \frac{\sigma_q}{\sigma_q - \sigma_{q+1}}$.

Hence,

$$\alpha \leq e^{\lambda h} \frac{\sigma_q}{\sigma_q - \sigma_{q+1}}.$$



Idealized method: analysis

Lemma (Linear bound [C., Gander, Vandereycken])

For an affine problem, if $\beta < 1$, the error verifies the linear bound

$$\max_{n \geq 0} \|E_n^k\| \leq \frac{\alpha^k}{1 - \beta} \max_{n \geq 0} \|E_n^0\|.$$

Lemma (Superlinear bound [C., Gander, Vandereycken])

For an affine problem, if $\beta < 1$, the error verifies the superlinear bound

$$\max_{n \geq 0} \|E_n^k\| \leq \frac{\alpha^k}{k!} \prod_{l=1}^k (N - l) \max_{n \geq 0} \|E_n^0\|.$$



Idealized method: analysis

Lemma (Optimal bound [C., Gander, Vandereycken])

For an affine problem, if $\beta < 1$, the error verifies

$$\|E_n^k\| \leq \alpha^k \sum_{i=0}^{n-k-1} \binom{k-1+i}{i} \beta^i \|E_{n-k-i}^0\|.$$

Proof ideas

- Extension of the proof proposed by [Gander and Hairer, 2008].
- Use the hypothesis $\beta < 1$.
- No loss of information in the last bound.



Idealized method: numerical results

Lyapunov equation

$$\dot{X}(t) = AX(t) + X(t)A^T + CC^T,$$

where

- $t \in [0, T]$
- $X(0) = X_0$ is low-rank.
- $A \in \mathbb{R}^{n \times n}$ is sparse.
- $C \in \mathbb{R}^{n \times \ell}$ is a tall matrix.

If A is the 1D discrete Laplacian, this equation models a 2D heat problem.

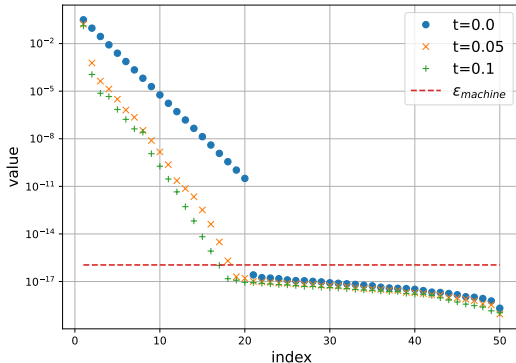


Figure: Singular values of the reference solution.



Idealized method: numerical results

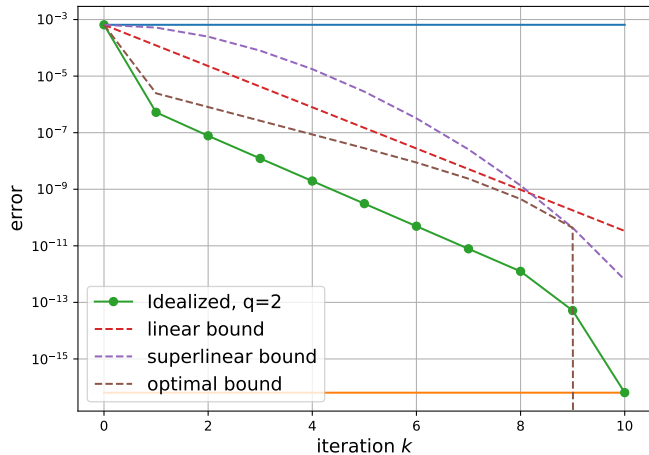


Figure: Lyapunov - Idealized algorithm with coarse rank $q = 2$. $\alpha \approx 0.16$, $\beta \approx 0.145$.



Idealized method: numerical results

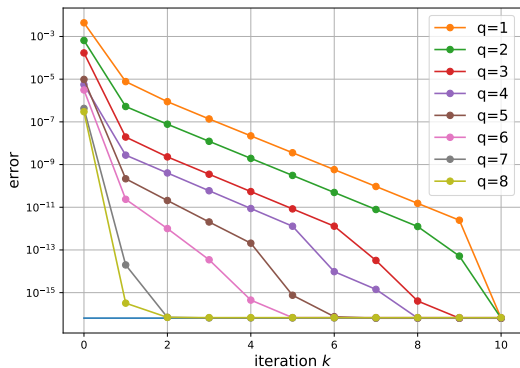


Figure: Lyapunov - Idealized algorithm for several coarse ranks q .

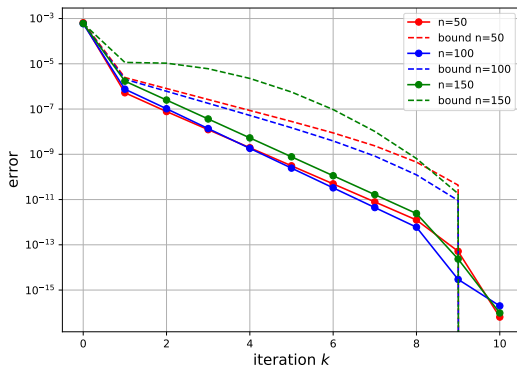


Figure: Lyapunov - Idealized algorithm for several sizes n .



Practical method: definition

Original problem

$$\begin{aligned}\dot{X}(t) &= F(X(t)), \quad t \in [0, T] \\ X(0) &= X_0.\end{aligned}$$

DLRA ([Koch and Lubich, 2007])

$$\begin{aligned}\dot{Y}(t) &= \mathcal{P}_{Y(t)} [F(Y(t))], \quad t \in [0, T] \\ Y(0) &= Y_0 \in \mathcal{M}_r.\end{aligned}$$

Definition (Low rank Parareal, practical formulation [C., Gander, Vandereycken])

For two ranks q, r such that $q < r$, we iterate

$$\text{(Initial value)} \quad U_0^k = Y_0$$

$$\text{(Initial iteration)} \quad U_{n+1}^0 = \psi_q^h \circ \mathcal{T}_q(U_n^0)$$

$$\text{(Iteration)} \quad U_{n+1}^{k+1} = \psi_r^h \circ \mathcal{T}_r(U_n^k) + \psi_q^h \circ \mathcal{T}_q(U_n^{k+1}) - \psi_q^h \circ \mathcal{T}_q(U_n^k)$$

where ψ_r^h is the exact flow of the DLRA with rank r . In particular, the rank of each iteration is at most $r + 2q$.



Practical method: comments

Remarks

- In practice, ψ_r^h and ψ_q^h are approximated by a DLRA integrator*.
- Problem solved by the coarse integrator $\neq$ Problem solved by the fine integrator.
- No analysis for the practical formulation yet.
- Same good numerical behavior as the idealized formulation!

DLRA integrators*	Order	Reference
KSL - Splitting projector integrator (2013)	1, 2	[Lubich and Oseledets,]
Projected methods (2019)	1, 2, 3	[Kieri and Vandereycken, 2019]
Splitting for stiff problems (2019)	1, 2	[Ostermann et al., 2019]
KLS - Unconventional splitting (2020)	1	[Ceruti and Lubich,]



Practical method: numerical results

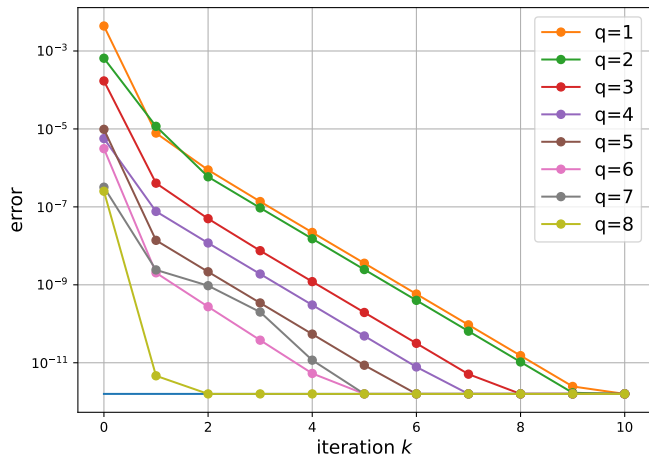


Figure: Lyapunov $\dot{X}(t) = AX(t) + X(t)A^T + CC^T$ - Diffusive problem - Practical algorithm (KSL).



Practical method: cost

Performance

- Coarse solver cost: $O(nq^2)$
- Fine solver cost: $O(nr^2)$
- Good potential speed-up since $q < r$.
- In our example: $q = 2$ and $r = 20$, the fine solver is about $\times 100$ more expensive.



Conclusion

Summary

- 1 Parareal can be applied in the context of DLRA.
- 2 Idealized algorithm
 - Satisfying theoretical bounds.
 - Good numerical results.
- 3 Practical algorithm
 - Efficient reformulation of the idealized algorithm.
 - Similar good results as the idealized algorithm.
 - No theoretical bound yet.

In the future

- Extension of the analysis to the practical algorithm.
- Understand better the role of the ranks in the convergence.



References

-  Breiding, P. and Vannieuwenhoven, N. (2021). Sensitivity of low-rank matrix recovery. *arXiv:2103.00531 [cs, math]*.
arXiv: 2103.00531.
-  Ceruti, G. and Lubich, C.
An unconventional robust integrator for dynamical low-rank approximation.
page 19.
-  Gander, M. J. and Hairer, E. (2008). Nonlinear Convergence Analysis for the Parareal Algorithm.
In Barth, T. J., Griebel, M., Keyes, D. E., Nieminen, R. M., Roose, D., Schlick, T., Langer, U., Discacciati, M., Keyes, D. E., Widlund, O. B., and Zulehner, W., editors, *Domain Decomposition Methods in Science and Engineering XVII*, volume 60, pages 45–56. Springer Berlin Heidelberg, Berlin, Heidelberg. Series Title: Lecture Notes in Computational Science and Engineering.
-  Kieri, E., Lubich, C., and Walach, H.
Discretized Dynamical Low-Rank Approximation in the Presence of Small Singular Values.
page 19.
-  Kieri, E. and Vandereycken, B. (2019). Projection Methods for Dynamical Low-Rank Approximation of High-Dimensional Problems. *Computational Methods in Applied Mathematics*, 19(1):73–92.
-  Koch, O. and Lubich, C. (2007). Dynamical Low-Rank Approximation. *SIAM J. Matrix Anal. & Appl.*, 29(2):434–454.
-  Lions, J.-L., Maday, Y., and Turinici, G. (2001). Résolution d’EDP par un schéma en temps pararéel. *Comptes Rendus de l’Académie des Sciences - Series I - Mathematics*, 332(7):661–668.
-  Lubich, C. and Oseledets, I. V.
A projector-splitting integrator for dynamical low-rank approximation.
page 18.
-  Ostermann, A., Piazzola, C., and Walach, H. (2019). Convergence of a Low-Rank Lie–Trotter Splitting for Stiff Matrix Differential Equations. *SIAM J. Numer. Anal.*, 57(4):1947–1966.



Appendix: Dynamical Low-Rank Approximation

Conditions for DLRA [Kieri and Vandereycken, 2019], [Koch and Lubich, 2007]

- F is a low operator rank (cheaper to evaluate on a low-rank matrix).
- F is Lipschitz: $\|F(X) - F(Y)\| \leq L \|X - Y\|$.
- F is one-sided Lipschitz: $\langle X - Y, F(X) - F(Y) \rangle \leq l \|X - Y\|^2$.
- F maps to a tangent bundle of $\mathcal{M}_r$: $\|F(Y) - \mathcal{P}_Y F(Y)\| \leq \epsilon$.

In practice, it works very well.

Theorem (Accuracy of DLRA)

Under the three assumptions above, the DLRA verifies

$$\left\| \psi_r^h(Y_0) - \phi^h(X_0) \right\| \leq e^{lt} \|Y_0 - X_0\| + \epsilon \int_0^t e^{ls} ds, \quad (1)$$

where ψ_r^h is the flow of the DLRA, and ϕ^h is the flow of the original problem.